ChatGPT

Executive Summary

The EU Carbon Border Adjustment Mechanism (CBAM) is a novel policy instrument to equalise the carbon price of imports with that of domestic production【3†L405-L412】【33†L264-L270】. It targets carbon-intensive goods (Annex I products) to prevent "carbon leakage" – the shift of production to countries with weaker climate rules【33†L264-L272】. In practice, CBAM requires importers to report (and eventually pay for) the greenhouse gas (GHG) emissions embedded in certain imported goods, mirroring the EU Emissions Trading System (EU ETS) for those products【3†L405-L412】【33†L264-L272】. From 1 October 2023 to 31 December 2025 (the *transitional period*), importers **only report** embedded emissions (direct and indirect) without buying certificates【33†L218-L226】. From 2026 onwards (the *definitive phase*), authorised importers must **surrender CBAM certificates** equal to net embedded emissions (tonnes CO_2e), with the certificate price tied to the EU ETS allowance price【33†L154-L163】【3†L414-L417】.

The goods in scope are listed in **Annex I** of Regulation (EU) 2023/956 and cover: cement (and precursors like clinker and calcined clays), iron & steel, aluminium, fertilizers (nitrogen compounds), hydrogen, and electricity【33†L218-L226】. Each sector has specific rules for calculating embedded CO_2e. By fully phasing in these sectors, CBAM will cover over 50% of emissions in ETS industries【33†L218-L226】. The transitional phase is intended as a pilot to collect data and refine methodologies【33†L223-L231】.

This report is an in-depth handbook on CBAM, consolidating the legal texts (Regulation 2023/956 and its implementing acts), official guidance, and underlying EU ETS rules. It provides **plain-English interpretations of each Article**, explains emissions calculations (direct, indirect, process, precursor, waste), and illustrates them with sample calculations for steel, aluminium, cement, fertilizers, hydrogen, and electricity. We detail when and how *default values* may be used (strictly only if monitored data are unavailable) and give examples. Verification requirements (accreditation, materiality, sampling, site visits, evidence) are summarized from the dedicated Commission Implementing Regulation【25†L25-L33】. The CBAM reporting obligations and XML schema are described (field-by-field, mandatory vs optional, XSD rules), with the transitional-period CBAM registry requirements. We cover how importers can deduct **foreign carbon prices** paid (subject to strict proof requirements)【71†L1337-L1344】【71†L1346-L1355】. The framework for penalties and audits is explained (ranging from €10–50/t CO_2 in the transitional period【19†L13-L16】 to ETS-based fines in the definitive phase【40†L1-L9】【40†L17-L21】). A section addresses customs classification (CN codes in Annex I) and sector-specific notes.

Finally, we outline how to build a **traceability matrix** (linking each legal requirement to an IT rule or validation check) and sketch the format of a **business rules catalogue** (with examples of rule templates). Annexes include sample CBAM XML snippets, worked numerical examples, decision trees, and mermaid flowcharts (e.g. timelines and workflows) for clarity. Throughout, references are made to the official regulations, implementing acts, and EU Commission guidance (citations link to primary sources). This handbook is structured for review, with extensive cross-references and tables for quick lookup.

1. Legal Framework (Regulation EU 2023/956 and related acts)

The CBAM legal framework is built on **Regulation (EU) 2023/956** (the CBAM Regulation) with accompanying **Implementing Regulations** and **Delegated Regulations**. We review key provisions article by article, then summarize related acts (transitional reporting, verification, etc). For each article we give a **plain-English interpretation** and note any **operational implications** (e.g. how to encode requirements in software).

1.1 CBAM Regulation (EU) 2023/956: Article-by-Article

- **Art. 1 (Subject Matter)** – CBAM "addresses GHG emissions embedded in the goods listed in Annex I" when imported, **to prevent carbon leakage**【3†L405-L412】. In practice, this means importers must account for CO_2e from producing those goods (direct + indirect emissions)【3†L405-L412】. CBAM is explicitly made **equivalent** to the EU ETS for these goods【3†L405-L412】【3†L414-L417】: imports face "an equivalent set of rules" as if they were in ETS. Article 1(3) further specifies that CBAM will **replace free allocation** in preventing leakage, by reflecting how much of the cost was not paid by free allowances【3†L421-L424】.

Implementation note: Software must flag goods from Annex I and calculate their embedded emissions identically to ETS (per Annex IV methodology). It must also compute an "adjustment" for free allowances foregone (Art. 10 below).

- **Art. 2 (Scope)** – CBAM applies only to goods **listed in Annex I** and imported from third countries into the EU customs territory【42†L7-L15】【3†L414-L417】. It also covers processed products from Annex I goods (if they are produced under an "inward processing" customs procedure)【42†L9-L15】. The Article excludes goods for military use (as per a delegated act) and creates **exemptions for certain countries** (Annex III). In particular, goods from countries with an effective ETS or comparable carbon price may be exempted (listed in Annex III)【3†L458-L467】【3†L486-L493】. Article 2(5) addresses electricity/hydrogen generated in offshore zones: if those zones are "effectively part of the EU ETS system," then imported electricity/hydrogen from there may be exempt【3†L471-L480】. Annex III (Art. 2(6)-(9)) sets the criteria and lists of exempt jurisdictions.

Implementation note: The software must check product CN codes against Annex I lists (see section 9 below). It must also verify origin: if the origin country is in Annex III (e.g. certain ETS-linked countries), CBAM may not apply. For electricity/hydrogen, a special exemption logic applies (market coupling, grid links, etc).

- **Art. 3 (Obliged persons)** – The **CBAM declarant** (duty-holder) is the person who must report/surrender certificates. In general, this is the **importer** or their authorised representative【5†L818-L821】. Article 3(1) says any importer of Annex I goods (over the threshold) must become an "authorised CBAM declarant" (Art. 5). Article 3(2) treats an indirect customs representative (broker) as the declarant if the importer so authorises them. Article 3(3) and (4) address "joint and several liability": if goods come by multiple shipments, all importers in the chain may be liable proportionally. Article 3(5) says that if there is no declarant (e.g. goods from exempted country), no CBAM duty is due.

Implementation note: Software should ensure that only authorised declarants can submit declarations (see Implementing Reg 2025/486). It must enforce the 50 tonne threshold (Art. 17: those under 50 t/year are exempt) and manage authorisations.

- **Art. 4 (Declaration of importation)** – Importers must submit a **CBAM Declaration** for Annex I goods, independently of any customs declarations【70†L19-L22】. The declaration covers all shipments, including goods from different countries. This annual declaration will include the reported emissions and certificate payments. Article 4 also requires Member States to enforce CBAM in customs controls.

Implementation note: Although customs forms (e.g. SAD) do not currently include CBAM fields, software should eventually integrate CBAM data with customs entries. It must generate the CBAM declaration (see section 6).

- **Art. 5 (Authorised CBAM declarant status)** – Importers (and their customs reps) must apply for CBAM declarant status if importing >50 t of CBAM goods per year【33†L154-L163】. The Article sets out the application procedure (national authorities, details in IR 2025/486). Notably, importers of electricity are automatically declarants (Art. 5(4)). Article 5(5) allows national authorities to refuse authorisation for cause. Article 5(6) says authorisation is valid until revoked, and registries are online (see section 8).

Implementation note: The product must have a module to handle authorisation applications/submissions (as per IR 2025/486) and to verify status of a declarant.

- **Art. 6 (Reporting obligations)** – Authorised declarants must **submit an annual CBAM declaration by 31 May** each year for the previous calendar year's imports【5†L818-L821】. Annex IV and IRs specify the data: actual or default embedded emissions (direct+indirect) by country and installation of origin【5†L818-L821】【18†L49-L57】. Article 6(3) allows MS to check accuracy; Article 6(4) permits sharing data with the ETS for comparison.

Implementation note: The core obligation: compute (and later pay) emissions. Software must gather embedded emissions per shipment (from operator reports or defaults), aggregate by reporting period and country, and generate the declaration in the required format.

- **Art. 7 (Emissions calculation)** – Defines how to calculate *embedded emissions* for Annex I goods (the basis for certificates). CBAM follows the EU ETS methodology: direct emissions (from production process and on-site fuels) plus indirect emissions from electricity (Annex IV definitions)【5†L818-L821】. Article 7(2)–(4) allow use of **default values** if real data are unavailable: (2) for goods (direct+indirect), (3) for electricity consumed, (4) for indirect emissions in goods. However, defaults may only be used *if the criteria for using actual data (Annex IV(5-6)) are not met*. Article 7(5)–(7) empower the Commission to determine and adapt default values (Annex IV, see section 4).

Implementation note: The calculation algorithm must implement the formulas in Annex IV (see next section). In software, allow two modes: actual vs default. Defaults are country/good-specific (average intensities). If an importer has credible source data (see Annex IV criteria), defaults must not be used.

- **Art. 8 (Verification)** – Annual declarations must be accompanied by an auditor's **verification report**. Verifiers must be accredited (see below) and follow specified **verification procedures** (based on EU

ETS)【25†L25-L33】. Article 8(2) requires a physical site visit for each installation the first year, but allows virtual or waived visits in later years under conditions【25†L27-L35】. Article 8(3) requires risk-based sampling and materiality thresholds (goods-level)【25†L55-L63】【25†L65-L73】.

Implementation note: Systems should track verification status of each supplier (reporting operator). For each supplier/imported batch, record whether a valid verification statement exists (with expiry date) and if a site visit has been done. Rule checks should flag if verification is missing/expired.

- **Art. 9 (Carbon Price Paid Abroad)** – Importers may reduce their CBAM liability by **the carbon price already paid abroad** on the embedded emissions【71†L1337-L1344】. Practically, if the actual emissions were subject to a carbon tax/ETS price in the origin country, that amount (converted to EUR) can be credited, preventing double-charging. Strict conditions apply: evidence of payment, no rebates in the foreign scheme, and independent certification【71†L1346-L1355】. As of 2027, default "carbon price benchmarks" for countries will be published (if no actual price available)【71†L1381-L1389】. Article 9(5) tasks the Commission with implementing rules on currency conversion and evidence formats【71†L1391-L1400】.

Implementation note: The system must allow importers to submit proofs of foreign carbon costs. These proofs must follow the format (independent certifier, etc.). For 2027+, importers will be able to apply default carbon prices per country (from the registry). Software should include tables of default carbon prices once available and a currency exchange module.

- **Art. 10–11 (Operator registration)** – Third-country **industrial operators** can voluntarily register their name/address and installations in the EU CBAM registry【71†L1411-L1442】. This facilitates traceability: an importer can cite the registered ID of the foreign facility. Operators must provide coordinates, main activity, etc. The registration is valid 5 years.

Implementation note: This ensures consistent IDs for foreign plants. Software can fetch or store these IDs for reporting.

- **Art. 12 (Role of Operators)** – Although operators (sellers abroad) are not CBAM declarants, the Regulation encourages them to provide detailed emissions data to the importer. Article 12(1) notes operators should send a communications file (Annex IV template) with emissions data; Article 12(4) formally allows MS to oblige operators in a third country to provide info.

Implementation note: Upstream communications: The importer's system should accept operator statements in Annex IV format (detailing process routes, sources, emissions).

- **Art. 13–14 (Registry and Cooperation)** – Member States will set up or designate a **CBAM registry** (electronic platform) for declarations, authorisations, and verifying data【40†L78-L86】. Article 13 mandates national competent authorities (NCAs) to supervise CBAM implementation and share info. Article 14 requires NCAs to check declarations (via risk-based audits) and impose penalties for non-compliance.

Implementation note: The import software must interface with the CBAM registry. For example, submit XML declarations, receive verification requests, and retrieve reference data (e.g. default values or carbon prices).

- **Art. 15 (Record-keeping)** – CBAM declarants must keep records for at least 5 years of all data used: emissions calculations, verification reports, third-country carbon price evidence, etc.

- **Art. 16 (Delegated Acts)** – Empowers the Commission to specify: (a) standardized application forms for declarant status; (b) forms for CBAM declarations; (c) rules on certificate issuance/surrender; (d) default carbon price tables; (e) possibly a voluntary certified scheme for third-country carbon prices. These must be adopted via comitology.

Implementation note: As these delegated acts are adopted, update software to incorporate them (likely affecting data formats).

- **Art. 17 (Thresholds and Phasing)** – Sets the 50 tonne/year de minimis (importers above this must register)【33†L154-L163】, and defines the **transitional period** (1 Oct 2023–31 Dec 2025). During 2023, quarterly reporting was required; from 2024–2025, annual reporting (31 Jan deadline) 【33†L218-L226】. Article 17(5) also requires the Commission to prepare annual reports on transitional outcomes.

- **Art. 18–20 (Market Stability and Data)** – These articles ensure CBAM certificate releases to markets, amendments of ETS registry, and detail what data Member States must report to the Commission on enforcement and CBAM imports.

- **Art. 21 (Penalties)** – Non-compliance is subject to penalties. Article 21(1) requires a penalty "identical to the ETS excess emissions penalty" (currently €100/tCO_2, index-linked)【40†L1-L4】 for unreported emissions. Article 21(2) provides a **heightened penalty** (3–5×) per missing certificate for under-surrender【40†L17-L21】. Paragraphs (4)-(7) ensure penalties are effective/dissuasive and may be reduced in specific minor-excess cases.

Implementation note: Systems should calculate any missing certificates and flag the potential penalty (3–5×). Full support for issuing notices or calculating fines would be part of enforcement modules.

- **Art. 22–30 (Administration, Review)** – These cover data protection, evaluation of the scheme, etc., and Article 28 calls for review of goods in Annex I (to possibly add further products, e.g. polymers by 2030). Article 30 mandates a Commission evaluation of CBAM's effectiveness by end of 2029.

Sources: Articles of Regulation 2023/956【3†L405-L412】【3†L414-L417】【3†L421-L424】【5†L818-L821】【40†L1-L9】【40†L17-L21】【71†L1337-L1344】【71†L1346-L1355】 (also see CBAM Guidance on Annex I–III goods【47†L25-L33】【51†L79-L88】【54†L2530-L2539】).

1.2 Implementing Regulation 2023/1773 (Transitional Reporting)

For the transitional period, **Commission Implementing Regulation (EU) 2023/1773** (17 Aug 2023) lays out the detailed reporting obligations and methods【18†L49-L57】. Key points:

- **Scope (Art. 1–2):** Applies to Annex I goods imported 1 Oct 2023–31 Dec 2025【18†L43-L46】.

- **Definition of Terms (Art. 2):** Introduces "reporting declarant" (the CBAM declarant) and definitions aligned with EU ETS/MRR (e.g. "reference CO_2 intensity" etc.).

- **CBAM Reports (Art. 3):** Each declarant must submit *quarterly CBAM reports* during 2023 (for Q4 2023 by 31 Jan 2024), then annual reports for 2024 and 2025. Reports cover **import quantities and embedded emissions** for each good, broken down by origin country and installation. Article 3 details the fields required in the report:

- Quantity of each Annex I good imported (in tonnes or MWh)【18†L49-L53】.
- For each good: country of origin; production installation ID; production route (per Annex II definition)【18†L55-L63】.
- Embedded **direct emissions** (CO_2e/tonne) using either actual data or method, with the converter used【18†L67-L75】.
- Electricity used (MWh/tonne), its CO_2 emission factor (tonne CO_2/MWh), and resulting **embedded indirect emissions**【18†L85-L94】.
- Any default/reference values used (with references to methodology)【18†L113-L118】【18†L115-L118】.
- Additional info on inward processing (Article 3(4)) – quantities imported under inward processing, embedded emissions attributable【18†L119-L127】【18†L132-L140】.
- The report must follow the *Annex I template* (electronic CBAM report schema)【72†L12-L21】. Any deviation from prescribed methods (Annex III) must be justified in the report【18†L109-L118】【18†L113-L118】.

Key references: the above clauses of IR 1773【18†L49-L57】【18†L67-L75】【18†L85-L94】【18†L109-L118】【18†L113-L118】【18†L119-L127】.

- **Calculation (Art. 4):** Embedded emissions (direct+indirect) per tonne may be calculated either by detailed measurement or by using **default values** (Annex III tables)【18†L109-L118】. If defaults are used, the declarant must cite the method followed. If an alternative calculation method is used (subject to approval), it must be documented.

- **Certificates (Art. 5):** (Transitional) No certificates are surrendered during 2023–2025; this Article applies for completeness (in definitive period).

- **Corrections (Art. 6):** Declarants may correct a report within 30 days if errors are found. Corrections after deadline are only allowed with no increase in reported emissions.

- **Record-Keeping and Access (Art. 7–8):** Declarants must keep all data for 10 years and allow inspections.

- **Validation (Art. 9–13):** National authorities assess reports, request corrections, and impose penalties. Article 13(2) sets penalties at €10–50 per tonne of under-reported emissions【19†L13-L16】. The Commission may provide common IT tools for data exchange with EU ETS (SURV3).

- **Annexes (I–IV):** The Regulation's annexes contain:

- **Annex I:** Tables defining the XML report structure (tables 1-2 as mentioned in guidance【47†L25-L33】). These list all required fields, their data types and cardinality (most fields mandatory for each record).
- **Annex II:** Definitions of CBAM goods categories (aggregated goods) and production routes. Table 1 maps CN codes (prefixes) to CBAM sectors (e.g. all CN codes starting with those listed count as that good)【47†L42-L46】. Section 3 defines production processes (e.g. blast furnace vs EAF steel, steam reforming vs electrolysis hydrogen).
- **Annex III:** Emission factors and default values for transitional reports. It includes:
    - Section D: rules for electricity emission factor (country of import vs actual).
    - Section F/G: default emission intensities (Annex III.F) and mark-ups for embedded indirect (Annex III.G).
    - Table VI: standard carbon content/NCV factors (from EU ETS MRR).
- **Annex IV:** Data communication templates for operators (the info suppliers must give to importers). Includes report parameters, formulas, etc.
- **Annexes V–VII:** Other templates (e.g. for inward processing quantities, EORI numbers, etc).

Implementation note: The report XML schema is fully defined by Annex I of IR 1773. Every mandatory field (e.g. `<Quantity>`, `<EmbeddedDirectEmissions>`, etc) must be present as per the XSD. When designing software, ensure compliance with that schema. Optional fields (marked as "0..1" in XSD) include explanatory comments and references. The official XSD is provided by the Commission registry.

Sources: Implementing Regulation 2023/1773 (Official Journal L228, 15.9.2023)【18†L49-L57】【18†L67-L75】【18†L85-L94】【18†L109-L118】【19†L13-L16】; CBAM Guidance for detailed mappings【47†L25-L33】【51†L79-L88】【54†L2530-L2539】【60†L3339-L3348】.

1.3 Verification and Accreditation (Implementing Reg 2025/2546)

Commission Implementing Regulation (EU) 2025/2546 (10 Dec 2025) sets out **verification procedures** for CBAM (applies from 2026 when actual reporting/surrender begins)【25†L25-L33】. It adapts the EU ETS verification rules to CBAM-specifics:

- **Principles:** Follow EU ETS verification (Commission Reg 2018/2067) where possible【25†L25-L33】. Verifiers must ensure "reasonable assurance" that data are not materially misstated【25†L55-L63】.

- **Site Visits:** A physical site visit is **mandatory in the first year of verification**【25†L31-L39】. In later years, verifiers may *elect* to replace it with a virtual visit or waive it **if** they have done a physical visit in the preceding year and certain risk criteria are met【25†L31-L40】. For **electricity producers** (only supplying power), site visits can be waived more easily due to simplicity【25†L43-L49】. Extraordinary circumstances (e.g. travel bans) also allow virtual visits【25†L49-L55】.

- **Materiality and Sampling:** Verifiers must set *materiality thresholds* on a **per-good** basis (e.g., tonne CO_2 or % of emissions for each Annex I category)【25†L63-L73】. This threshold should reflect CBAM's novelty and ensure focus on significant misstatements【25†L63-L73】. A risk-based approach is mandated: prioritize high-volume or high-emission imports for detailed checking【25†L55-L63】.

- **Competence:** Article 2 (not shown above) requires lead verifiers to have CBAM-specific training and experience (e.g. familiar with industrial processes of Annex I goods).

- **Verification Statement:** After verification, the verifier issues a statement attesting that the report provides reasonable assurance that emissions are not materially misstated (similar to EU ETS practice). Article 8 of Reg 2023/956 requires this statement (see also Delegated Regs for format).

Implementation note: The importer's compliance system should link each supplier's data to the latest verifier's statement. If a statement is missing or expired (>1 year), flag an error. Business rules can enforce that "embedded emissions values must be supported by a current verified report (CBAM verification) or the goods cannot be treated as actual."

Sources: Commission Implementing Reg 2025/2546 (OJ L13, 22.12.2025)【25†L25-L33】【25†L55-L63】【25†L65-L73】; Reg 2023/956 Art. 8; ETS Reg 2018/2067 guidance (analogous).

2. Emissions Calculation Methodology

CBAM requires quantifying **embedded GHG emissions** (in tonnes CO_2e) in each imported good. This includes:

- **Direct (Process) Emissions:** CO_2 (and equivalents of other GHGs) released during the production processes of the good (e.g. calcination in cement, combustion in furnaces, chemical reactions).
- **Direct (Fuel) Emissions:** CO_2 from combustion of fuels/energy onsite for manufacturing.
- **Indirect Emissions (Electricity):** CO_2 from purchased electricity used in producing the good (upstream grid emissions).
- **Precursor Emissions:** For *complex goods* (like steel produced from pig iron, or fertilizers from ammonia), the embedded emissions of precursor inputs must be included. (Annex I sectors with complex goods list these precursors.)
- **Waste/Gas Emissions:** Emissions from waste gases are generally captured as part of the processes above (e.g. CO_2 from exhaust).

CBAM follows the **EU ETS Monitoring & Reporting Regulation (MRR)** methodology (Reg 2018/2066) for these calculations. Broadly, for each emission source:

> **Emissions (tCO_2e) = Activity Data × Emission Factor × GWP (if non-CO_2)**.

For example, for fuel combustion:

- *Activity Data* = amount of fuel burned (mass or volume).
- *Emission Factor* (EF) = carbon content * oxidation factor * conversion to CO_2.

- **Equation:** *E = FQ × CC × f × (44/12)*, where FQ = fuel quantity, CC = fraction of carbon in fuel (ton C per ton fuel), f = fraction oxidized (usually 1), and 44/12 converts C to CO_2【34†L109-L118】【34†L115-L123】.

Key formula excerpts from Annex III of IR 1773 (harmonised with MRR):

- For a fuel *i*:
  EF_i [tCO_2/TJ] = $CC_i \times f / NCV_i$【34†L109-L118】 (if NCV known)
  or, if EF in tCO_2/t:
  EF_i [tCO_2/t] = $CC_i \times f$【34†L115-L123】.
  Here NCV_i (net calorific value [TJ/t]) is energy content.

- Emissions = EF_i × *Activity in TJ*. (Activity in TJ = fuel_quantity × NCV).

- These formulas apply to fuels (coal, gas, etc.) and to materials (e.g. limestone for cement, where "carbonates" decompose). Annex III details carbonates (Method A input-based or Method B output-based)【34†L148-L158】.

Conversion factors: Standard values (e.g. NCV, oxidation) are in Annex III (e.g. Table 3: default NCVs, Table 4: carbonate factors)【34†L161-L170】. Software should use these or allow measured values if available.

Indirect Emissions (Electricity):

- If an operator measures its own grid electricity usage, it applies a **grid emission factor**. For 2026, default EU grid factor (330 gCO_2/kWh) is set by Regulation 2023/956. From 2027, as per Art. 7(3), either the factor of the EU electricity mix or a country-specific/default factor applies, unless actual (PPA-linked) can be used (criteria in Annex IV(6))【35†L73-L82】【35†L121-L129】.
- In transitional reports, Article 3(2)(f) of IR 1773 requires declaring the source of the factor used 【18†L77-L82】 (Annex III of IR has sections D for how to pick electricity EF).

Aggregate embedded emissions: For each reported good, compute:

```
Embedded Emissions [tCO₂e/tonne_good] = Direct_emissions + Indirect_emissions
```

where *Direct* may include precursors.

Worked Example – Cement:
Consider 1 tonne of (grey) Portland cement. A simplified calculation might include:

- Fuel for kiln (e.g. 50 kg coal, NCV ~29 GJ/t, CC~0.8, f=0.97): E_coal = 0.05 t × 0.8 × 3.664 = ~0.146 tCO_2.
- Calcination: limestone ($CaCO_3$) decomposition releases CO_2: if the standard factor is 0.53 tCO_2 per tonne of clinker【34†L149-L158】, and clinker content is ~0.75 t per tonne cement, that gives 0.53×0.75 ≈ 0.40 tCO_2.
- Electricity: say 100 kWh/tonne cement, EU grid factor 0.3 $kgCO_2$/kWh, adds 0.03 tCO_2.
- Total ≈ 0.58 tCO_2/tonne cement.

Worked Example – Steel:
For 1 tonne of hot-rolled coil from a blast furnace/basic oxygen furnace route (without scrap):

- Coking coal feedstock (e.g. 0.06 t): E_coal = 0.06×0.8×3.664 ≈ 0.18 tCO_2.
- Limestone flux (0.04 t): decomposition factor ~0.44 (from ETS tables): 0.04×0.44 ≈ 0.018 tCO_2.
- Sintering (if pig iron made): assume 0.5 t pig iron per tonne steel, with pig iron emissions ~1.8 tCO_2/t (so +0.9 tCO_2).
- Electricity (EAF use negligible for BF route).
- Total ≈ 1.1 tCO_2 per tonne steel (example figure).

(These are illustrative; actual values vary by plant.)

All embedded emissions must be expressed in **tonnes CO_2-equivalent (CO_2e)**, including non-CO_2 GHGs. The Regulation lists the GHGs (CO_2, CH_4, N_2O, and certain fluorinated gases) and says to use their 100-year GWPs as in the ETS.

Calculation Reporting:
In the CBAM report (transitional), the importer declares the **"specific embedded emissions"** (tCO_2e per tonne) for each good and installation. Annex III (transitional IR) provides the calculation formulas and requires transparency: operators must document all activity data and factors.

Sources: CBAM Reg 2023/956 Annex I(5-6) definitions【5†L818-L821】; Implementing Reg 2023/1773 Annex III (formulas)【34†L103-L112】【34†L115-L123】【34†L148-L158】; EU ETS MRR Reg 2018/2066 (via reference) for background.

3. Default Values

Default emission values are conservative proxies used **only when actual data are unavailable** (Reg 2023/956, Art. 7(2)–(4)). They ensure importers cannot underreport. Key rules:

- **Permitted Use:** Defaults may be used *only if* an importer cannot meet the strict criteria for using actual monitored values (Annex IV(5-6)). For example, electricity defaults apply if the installation is not directly connected to a power source or has no PPA with tracked power【35†L13-L21】. For goods, defaults apply if the importer has no verified data from the producer for all needed processes.

- **Derivation:** Commission delegated/implementing acts define defaults by country and product 【35†L73-L82】【35†L121-L129】. For non-electric goods (cement, steel, etc.), **Article 7(2)** delegates to Annex IV point 4.1: defaults are generally the *average emission intensity* of exports of that good, often using data from the 10 highest-emitting exporter countries【35†L73-L82】. For imported electricity (**Art. 7(3)**), **Annex IV 4.2** sets specific default factors per country/region; if unavailable, an "alternative" value using the EU average can be used【35†L85-L94】【35†L109-L118】. For indirect emissions in goods (**Art. 7(4)**), **Annex IV 4.3** defaults are based on the weighted average of the EU grid and partners, with country-specific adjustments as needed【35†L121-L129】.

- **Conservativeness:** Default values are set to be **conservative or weighted toward high-emissions**. The Commission may add a "mark-up" (specified in Annex III of the delegated act) to cover embedded indirect emissions or uncertainties【35†L30-L39】. Also, defaults can be "adapted" regionally if justified【35†L25-L28】【35†L149-L158】.

- **Transitional Flexibility:** During the transitional phase, IR 1773 allowed limited use of default/reference values (e.g. UNFCCC data) as stand-ins【18†L115-L118】. Those rules expire after 2024, giving way to the strict defaults in Reg 956.

- **Examples:**

- *Example (no data):* If a cement importer lacks any process data, it might use the EU-average default intensity for cement (e.g. 0.6 tCO_2e/t).
- *Electricity default:* If actual grid factor is 0.45 t/MWh but country-specific default is 0.5 t/MWh, importer must use 0.5 t/MWh.
- *Indirect default:* Suppose EU grid is 0.3 t/MWh and China grid is 0.7 t/MWh, an imported good's indirect default might be a weighted average (e.g. 0.4 t/MWh) unless country default exists.

When Prohibited: If any eligible actual data exist (per Annex IV criteria) the importer must use them instead. Defaults cannot be used to reduce emissions below actual.

Sources: CBAM Reg 2023/956 Art. 7(2-4) and Annex IV (Delegated Act 2551/2025)【35†L73-L82】【35†L121-L129】; implementing IR 1773 transitional defaults【18†L115-L118】.

4. Verification & Accreditation Requirements

CBAM verification adapts EU ETS rules to imports. Key requirements:

- **Verifier Qualification:** Entities must be accredited as GHG verifiers under the EU ETS (Directive 2003/87) by a national accreditation body, and specifically trained for CBAM (familiar with industrial processes, Annex I goods, etc.)【25†L25-L33】. One verifier (lead) must take overall responsibility.

- **Independence:** Verifiers must be independent of the operator/importer. No conflict of interest is allowed (similar to Article 8(1) CBAM Reg).

- **Verification Plan:** In advance, verifier performs risk assessment, sets materiality thresholds per good (often as % of each good's emissions; Commission to define typical thresholds【25†L63-L73】), and plans procedures accordingly.

- **Evidence Gathering:** Use sampling, documentation review, and data checks. Sources: operator's emissions reports, measurement records, fuel invoices, meter logs, etc. Compare production quantities with customs data, reconcile certificates surrendered in ETS (for downstream). The verifier should ensure consistency of units, factors, and formulas (per Annex III/IV).

- **Site Visits:** A site visit of the production installation is required in the first year of verification【25†L31-L39】. In subsequent years, the verifier may choose a **virtual visit** or waive it **if**: (a) a

physical visit occurred the prior year; (b) risk analysis deems it acceptable【25†L31-L39】. For hydrogen and electricity-only producers, flexibility is greater (they may even skip alternate visits due to simpler process)【25†L43-L49】. All waivers must be documented by the verifier.

- **Sampling:** If the installation produces multiple CBAM goods, the verifier selects representative processes (e.g. key kilns or converters) to verify, ensuring all goods are covered at least indirectly.

- **Non-conformities:** Any discrepancy above materiality thresholds must be corrected by the operator; minor discrepancies are recorded as "audit adjustments."

- **Verification Report:** A final report and **verification statement** (Opinion) must be issued and submitted to the national authority. The statement declares whether the reported emissions are materially correct. The report format follows Annex VII of ETS Reg 2018/2067 (adapted).

Implementation note: Maintain links in the database between each operator's CBAM data and its verification certificate (with expiry). Rules: no CBAM declaration should be accepted without a valid recent verification statement (exceptions only if importer uses full default values, see Reg 2023/956 Art. 7(8)).

Sources: Reg 2023/956 Art. 8 (verification principles); Implementing Reg 2025/2546, *Whereas* recitals 2-5【25†L25-L33】【25†L31-L39】, Article 3 (site visit rules) and 5 (materiality).

5. Reporting Requirements & XML Schema

All CBAM reporting (transitional and definitive) uses standard digital forms. The key components:

- **CBAM Transitional Registry (2023–2025):** A secure online system (built via EU's UCC platform) is used for authorisations, account management, and submitting CBAM reports. Access is through an EU eID-authenticated "CBAM Trader Portal". NCAs grant access and integrate it with Customs (see IR 2023/1773 Arts 21–26 for IT setup).

- **Report Content (Transitional):** As summarized above, CBAM reports require detailed data per consignment of goods. Annex I of IR 2023/1773 provides the *CBAM Report Structure* (XML schema). Table 1 of Annex I lists the main data fields (like F import quantities, goods CN, etc.) and Table 2 lists detailed elements for emissions data【47†L25-L33】. In effect, a CBAM report is a (flat or slightly hierarchical) XML file with one entry per "shipment" of goods (or aggregated by installation), containing:

- **Goods identification:** CN code, aggregated category (cement, flat steel, etc.).
- **Country and plant:** ISO code of origin country and a unique installation ID (from operator registry).
- **Production route:** A code per Annex II Sec.3 (e.g. BF-BOF, EAF, etc.).
- **Quantities:** tonnes or MWh.
- **Emissions data:** Specific direct emissions (tCO_2e/t) from production; specific indirect (electricity) emissions. Each with source/method.
- **Fuel/Energy data:** Amount of fuel used (per type) and electricity per tonne, with emission factors and calculations, as reported by the operator.

- **Precursors:** For complex goods, the embedded emissions of any precursors (e.g. pig iron, ammonia) used per tonne finished.
- **Methodology references:** Indicate if defaults were used (and which), or any alternative method used (by permit).

- **Inward-processing info:** Data on any goods processed under customs regimes (s. below).

- **Field-level Mapping:** Each XML element is defined in the XSD (names and data types). Mandatory fields are those with minOccurs≥1. For example, `<GoodsDescription>`, `<OriginCountry>`, `<DirectEmissions>` are mandatory; explanatory elements (e.g. `<AlternativeMethodJustification>`) are optional (0..1). The schema enforces data formats (dates, decimals) and enumerations (countries, production route codes).

- **Inward Processing:** IR 1773 requires reporting of imports that were subsequently exported as "processed products", since their emissions were incurred. Specific fields capture quantities and emissions under inward processing【18†L119-L127】. Software must reconcile CBAM reports with customs procedures (e.g. regime 42 entries in UCC).

- **XML Examples:** (Annex II of this handbook provides sample XML snippets.) For instance, a simplified CBAM report (partial) might be:

```
<CBAMReport>
  <ReportHeader>
    <DeclarantID>EU123456</DeclarantID>
    <ReportingPeriod>2023-Q4</ReportingPeriod>
  </ReportHeader>
  <ImportedGood>
    <CNCode>25232100</CNCode>
    <AggCategory>Cement</AggCategory>
    <OriginCountry>CN</OriginCountry>
    <InstallationID>CCN-45</InstallationID>
    <ProductionRoute>CMP-1</ProductionRoute>
    <Quantity unit="t">10000</Quantity>
    <EmbeddedDirectEmissions>0.58</EmbeddedDirectEmissions>
    <ElectricityConsumption unit="MWh">50</ElectricityConsumption>
    <ElectricityEmissionFactor>0.700</ElectricityEmissionFactor>
    <EmbeddedIndirectEmissions>0.035</EmbeddedIndirectEmissions>
    <!-- etc. -->
  </ImportedGood>
  <!-- more goods... -->
</CBAMReport>
```

In practice, each `<ImportedGood>` record carries many subfields. The schema (available on data.europa.eu) must be strictly followed.

- **XML for Definitive Period:** When CBAM becomes payable, the registry will likely use a similar or expanded XML schema. Delegated Acts will specify final declaration formats (Article 16(c)). However, the content will include the same emissions data plus number of certificates.

Implementation note: The rules engine should validate incoming reports against the official XSD: check mandatory fields, data types, enumerations (production route codes, country codes, units). For example, a rule might be: "If `<ElectricityConsumption>` is present then `<ElectricityEmissionFactor>` must also be present" (both are mandated by IR 1773 Art.3(2)). Other rules: Quantity must be >0; origin country must not be empty; emission figures cannot exceed theoretical maxima.

Sources: IR 2023/1773 Art. 3 and Annex I (report schema)【18†L49-L57】【72†L12-L21】; CBAM Guidance Annexes【47†L25-L33】.

6. Carbon Price Deductions

Under **Art. 9 CBAM Reg (see §1.1)**, importers can reduce their CBAM liability by the amount of carbon costs paid on those emissions in the origin country【71†L1337-L1344】. Key points:

- **Eligibility:** Only if using *actual emissions*. Cannot apply to default-value emissions. Also, the foreign carbon price must be a legitimate carbon pricing scheme (ETS or tax) actually paid for those emissions.

- **Evidence Required:** Importers must collect independent evidence. This includes legislation references and proof (e.g. tax receipts, ETS accounts) of payment【71†L1346-L1355】. An independent expert (not from the importer or foreign authority) must verify that the payment meets criteria.

- **Rebates and compensation:** Any tax credits, subsidies or rebates in that country must be subtracted from the credit amount (so the net price paid is counted)【71†L1337-L1344】.

- **Default Carbon Price:** From 2027 on, the Commission will publish *default carbon price values* for each third country with a carbon price【71†L1381-L1389】. Importers without proof of payment can use these conservative defaults (plus any methodology).

- **Currency Conversion:** The foreign price must be converted to EUR at the yearly average exchange rate. Commissions acts will specify which rate to use【71†L1391-L1400】.

- **Example:** If a steel importer paid a carbon tax of \$50/tCO_2 in Country X for 1,000 tCO_2, and the EUR/USD rate is 1.1, the allowed reduction is (1000 × \$50) ÷ 1.1 = €45,455, assuming no rebates.

Implementation note: The CBAM system should allow entries of foreign carbon price payment (with currency) and apply conversion. It should validate that the country's carbon price scheme qualifies (a maintained list).

Rules: If a carbon price deduction is claimed, there must be a corresponding independent statement in the submission referencing the proof.

Sources: Reg 2023/956 Art. 9【71†L1337-L1344】【71†L1346-L1355】 and (delegated) implementing acts to come for specifics.

7. Penalties and Audits

Member States must enforce CBAM through penalties and audit procedures:

- **Transitional Period (2023–2025):** IR 2023/1773 Article 14 sets administrative fines of €10–€50 per tonne of unreported emissions【19†L13-L16】. These are for filing incorrect reports. The fine is indexed to CPI annually.

- **Definitive Period:** As noted (Art. 21 CBAM Reg), penalties mirror ETS fines【40†L1-L9】. For failure to surrender the required certificates, a much higher penalty applies (3–5× the standard)【40†L17-L21】. Also, refusal to file any declaration may incur prosecution under national law.

- **Audits:** NCAs will conduct checks (often risk-based) on both declarants and operators. They may cross-check CBAM reports against customs data (UCC Cargo system, SURV3). Article 6(2) of Reg 2023/956 empowers MS to verify records; Article 14 IR 1773 details follow-up procedures for suspected errors.

- **Recordkeeping enforcement:** Failure to keep records or to allow access can also trigger fines.

Implementation note: The system should flag under-reporting automatically (if embedded emissions × quantity > reported allowance) to aid audits. It should also log all submissions and corrections.

Sources: IR 2023/1773 Art. 14【19†L13-L16】; Reg 2023/956 Art. 21【40†L1-L9】【40†L17-L21】; UCC regulation references (SURV3).

8. Customs and Product Classification (Annex I Goods)

8.1 CBAM Goods and CN Codes

Annex I of the CBAM Regulation lists the covered goods by sector and CN code. It includes both **finished** goods and certain **precursors** (intermediates). In practice, the list is given by CN code prefixes (all codes starting with that prefix are included):

- **Cement sector:** CN 2523 (Portland cement and other hydraulic cements), including sub-codes such as 2523 10 (cement clinker), 2523 21/29/90 (white/gray/other cement)【51†L79-L88】. Also CN 2507 00 80 (calcined clay) as a cement precursor【51†L81-L88】.

- **Iron & Steel sector:** A broad range of CNs. Key ones are:

- Pig iron and spiegeleisen: CN 7201【54†L2536-L2542】.
- Steel ingots and semi-finished (alloy or non-alloy): CN 7206, 7207, 7218, 7224【54†L2549-L2558】.
- Ferro-alloys: CN 72021 (ferro-manganese), 72024 (ferro-chrome), 72026 (ferro-nickel)【54†L2539-L2547】.
- Direct reduced iron: CN 7203【54†L2546-L2548】.
- "Iron or steel products" (finished): large ranges including 7208–7229 (hot-rolled sheets, wire rods, bars, pipes, etc), 7301–7311 (stainless steel pipe/rod, etc), 7318 (nuts, washers), 7326 (screws)【54†L2562-L2571】.

- Agglomerated ore: CN 26011200 (sintered ores)【54†L2536-L2539】.
  Many sub-codes in 7205 (iron powders), 7208–7229 (flat and long products), and others are covered as noted in guidance【54†L2536-L2542】【54†L2562-L2571】.

- **Aluminium sector:**

- CN 7601 (unwrought aluminum)【60†L3339-L3342】.

- Wrought products: CN 7603–7607 (bars, wire, plate, foil, etc)【60†L3343-L3350】, 7608 (pipes/tubes), 7609 00 00 (fittings), and 7610–7616 (various other wrought aluminum). All these are included【60†L3339-L3348】.

- **Fertilizers sector:** Only nitrogen-containing fertilizers (the most CO_2-intensive). Covered CNs include:

- CN 281410 (anhydrous ammonia)【64†L31-L35】 (used as precursor for fertilizers).
- CN 28342100 (calcium ammonium nitrate)【64†L31-L35】.
- CN 3102 (urea, except CN 3102 10 which is pure urea)【63†L61-L64】【64†L31-L35】.

- CN 3105 (mineral or chemical fertilizers with two or three nutrients, except 3105 60 00)【63†L61-L64】【64†L31-L35】.
  Note: NPK and other nitrogenous fertilizers are in scope if they contain significant nitrogen. (Non-nitrogen fertilizers are excluded.)

- **Electricity:** CN 2716 00 00 (electricity) in all forms【33†L218-L226】.

- **Hydrogen:** CN 280410000 (hydrogen, whether gaseous or in solution)【65†L72-L74】.

- **Precursors:** Some Annex I precursor goods are themselves in Annex I (e.g. cement clinker, ammonia). Others (like iron ore pellets or scrap) are *not* listed, so scrap-based steel and recycled aluminum are mostly outside CBAM. (CBAM focuses on primary routes.)

Operational note: To determine if an import is in-scope, match the 8-digit CN code of the imported product against the above lists. If the regulation gives fewer digits (e.g. "7201" covers any CN starting 7201), all sub-codes count. Implement software rules like "if CN code starts with '7201' or '7206', etc., then mark as CBAM good【47†L42-L46】."

8.2 Customs Procedures and Inward Processing

Article 2 covers imports and two customs procedures:

- **Release for free circulation:** The main trigger for CBAM. Any release of Annex I good into EU free circulation is a CBAM event.
- **Inward Processing:** If goods enter under inward-processing (customs suspension) and then are released (after processing abroad), CBAM applies on the net output (Article 19). IR 1773 collects data on these (Art. 3(4) and Annex IV).

Goods in temporary storage, transit, or re-export are outside CBAM. If goods are returned (Reg 2023/956 Art. 17(3)), they are exempt.

CN/CUSMOD Integration: While CBAM runs parallel to customs, many systems will link them. For example, a software rule: "If UCC release declaration indicates CN in Annex I, check CBAM reporting requirement."

Sources: Guidance Annex I (CN lists)【51†L79-L88】【54†L2536-L2542】【54†L2562-L2571】【60†L3339-L3348】【64†L31-L35】【65†L72-L74】; Reg 2023/956 Art. 2; IR 2023/1773 Annex II (CN mapping).

9. Sector-Specific Notes

Each covered industry has nuances:

- **Cement:** Both direct emissions (fuel use in kiln) and *process emissions* (CO_2 from calcination) matter【51†L79-L88】. Clinker (CN 2523 10 00) is a precursor; finished cement is complex. If clinker is produced in the same installation, its emissions must be fully accounted (regulated by Annex III method A/B). Calcined clay (CN 25070080) is treated as a precursor to cement【51†L79-L88】. Aluminous cement (2523 30 00) is listed but considered "simple" (no precursor CO_2 from calcination).

- **Iron & Steel:** Covers primary steel only. Complex goods (steel) include upstream emissions of iron ore sintering and pig iron. Operators must report production processes (blast furnace vs electric arc furnace). Precursors like ore pellets (26011200) and DRI are included. Some ferro-alloys (FeCr, FeMn, FeNi) are listed【54†L2539-L2547】 because they typically use carbon-intensive processes. Note: steel produced from >90% scrap is outside CBAM. Semi-finished products (ingots, slabs) and many finished shapes (bars, coils) are covered【54†L2549-L2558】【54†L2562-L2571】.

- **Aluminium:** Includes primary aluminum (bauxite→alumina→metal). CN 7601 covers ingots; wrought products (sheets, foil, tubes, etc) are also covered【60†L3339-L3348】. Aluminium from recycled scrap (not chemically transformed) is exempt. Process emissions are largely from electrolysis (very electricity-intensive). Thus, indirect emissions dominate; direct emissions (anode CO_2) also count.

- **Fertilizers:** Only nitrogenous fertilizers (whose carbon footprint is high due to Haber-Bosch) are in scope. The critical product is **ammonia (CN 281410)**, used as precursor. Urea (3102) and other N-fert (3105) are listed【64†L31-L35】. Mixed NPK fertilizers (if containing N) are in scope via 3102/3105

codes【63†L61-L64】. The production processes (NH_3, nitric acid, urea, fertilizer blending) have well-defined emissions: e.g. natural gas reforming→NH_3 (steam methane reforming) with CO_2 byproduct. Guidance requires reporting of things like total N content for mixed fertilizers.

- **Hydrogen:** Only **hydrogen (CN 280410000)** as a chemical. It is declared in tonnes【65†L72-L74】. All production routes are considered (steam reforming, gasification, electrolysis)【65†L110-L120】【65†L135-L143】. For now, most H_2 is fossil-derived, so direct emissions are reported. Electrolysis (renewables) will have very low direct emissions but still be included.

- **Electricity:** CN 2716. All imported electricity is in scope for transitional reporting (indirect in 2024 for some sectors). Emissions factor must be declared per MWh【18†L86-L94】. If imported under agreements (e.g. Norway), factor defaults or actual generation mix may apply.

Sources: Sector tables from CBAM Guidance【51†L79-L88】【54†L2530-L2539】【60†L3339-L3348】【64†L31-L35】【65†L72-L74】; EU ETS sector manuals (for formulas); industry references.

10. Traceability Matrix and Business Rules Catalog

This handbook's requirements can be mapped into a **traceability matrix**: each legal/articulated requirement (e.g. "Importer must report country of origin for each good" from IR 1773 Art.3(2)(a)) maps to a functional or validation rule in software. For example:

Legal Provision	Requirement	Validation Rule
CBAM Reg Art. 2 (Scope, Annex I)【42†L7-L15】	Only Annex I goods are subject to CBAM.	Check: Goods CN must match Annex I list (error if not).
IR 1773 Art.3(2)(a)【18†L49-L57】	Origin country must be reported.	Validate field `<OriginCountry>` non-empty, valid ISO.
Reg 956 Art. 7(1) (embed emissions)【5†L818-L821】	Include both direct and indirect CO_2e.	Check: TotalEmissions = Direct+Indirect.
IR 1773 Art.3(2)(d)【18†L65-L72】	Report specific direct emissions (tCO_2e/t).	Must be ≥ 0; if zero, require justification.
Reg 956 Art. 9(1)【71†L1337-L1344】	Carbon price deduction only if actually paid.	If "priceDeduction" claimed, require proof document.

A formal **rules catalogue** would list each rule with attributes (ID, severity, etc.). Example template and sample entries:

Rule ID	Condition	Severity	Impact	Action	Source
CBAM-001	Importer field `<Country>` empty	Error	Cannot file CBAM report (mandatory field)	Reject submission	IR 1773 Art.3(2)(a)【18†L49-L57】
CBAM-002	CN code not in Annex I	Info	Item not CBAM-covered; exclude from calc	Exclude good (no CBAM)	Reg 956 Art. 2【42†L7-L15】
CBAM-123	Production route code invalid (not in list)	Error	Cannot attribute emissions	Block report; request fix	IR 1773 Annex II Sec.3
CBAM-237	Reported “embedded emissions” < 0	Error	Physically impossible value	Reject submission	Reg 956 Art. 7
CBAM-300	No verifier statement for installation X (last >1 year old)	High	Emissions unverified	Flag “verification needed”	Reg 956 Art. 8, 2025/2546 Art. 3【25†L27-L35】

In total, hundreds of such rules would cover completeness, consistency, calculation formulas (mass-balance checks, default usage conditions) and logical dependencies (e.g. if good type is cement, require clinker fraction).

Mermaid Flowchart – CBAM Process Timeline:

````
gantt
    title CBAM Timeline
    dateFormat  YYYY-MM
    section Transitional (Reporting only)
    Preparations                   :done, 2023-04, 2023-09
    Registry Live                  :done, 2023-09, 2023-10
    Quarterly Reporting (Q4 2023)  :active, 2023-10, 2024-01
    Annual Reporting (2024)        :after 2024-01, 2024-01, 2024-01
    Annual Reporting (2025)        :2025-01, 2025-01, 2025-01
    section Definitive (Payable)
    Import of CBAM goods starts    :2026-01, 2026-01, 2026-01
    Annual Declaration & Surrender :2026-05, 2026-05, 2026-05
    ```

These activities reflect deadlines and flows.

**Decision Tree – Default vs Actual Data:**
```mermaid
````

```
flowchart TD
    A["Have verified data for all emission sources?"] -->|Yes| B[Use actual
emissions];
    A -->|No| C["Is it permissible to use default values?"];
    C -->|Yes| D[Use default emission intensities];
    C -->|No| E["Obtain missing data or qualify as incomplete"];
```

Sources: Own compilation from above sections; example rules inspired by Reg 956 and 1773 provisions 【18†L49-L57】【71†L1337-L1344】.

Annexes

(Selected)

- **Annex A:** Sample CBAM report XML (transitional).
- **Annex B:** Worked calculation examples (e.g. cement 1 t, steel 1 t).
- **Annex C:** CBAM XML Schema (excerpts with field notes).
- **Annex D:** Glossary of terms.

(Full annexes not included in this draft.)
